
(a) Key/value cache bank
always readable
t_1–t_4
t_5–t_8
t_j readable iff $j \le i$
$\mathbf{k}_j$
$\mathbf{v}_j$
P_0
P_1
S_1
S_2
t_9
t_{10}
t_{11}
t_{12}
prefix
summaries
current chunk
(b) Summary availability
query steps
seal
t_9–t_{12}
q_9
q_{10}
q_{11}
q_{12}
S_3
$S_3 \notin \mathbf{K}$ for q_9–q_{12}
later reads
(c) Additive causal mask over the bank
prefix
summaries
local causal block
reads
q_9 0 0 0 0 0 −∞ −∞ −∞ 5
q_{10} 0 0 0 0 0 0 −∞ −∞ 6
q_{11} 0 0 0 0 0 0 0 −∞ 7
q_{12} 0 0 0 0 0 0 0 0 8
0 permitted
−∞ blocked
$j \le i$ edge
$\mathbf{A} = \mathrm{rowsoftmax}(\mathbf{Q}\mathbf{K}^\top/\sqrt{d_k} + \mathbf{M})$, $\mathbf{O} = \mathbf{A}\mathbf{V}$
Q
K^T
+M
softmax
A
V
O
$4 \times d_k$
$d_k \times 8$
4×8
$8 \times d_v$
$4 \times d_v$
$M_{ij} = 0$ if $j \in \{P_0, P_1, S_1, S_2\}$ or t_j with $j \le i$
$M_{ij} = -\infty$ otherwise
(d) Worked readout for query q_{10}
P_0 P_1 S_1 S_2 t_9 t_{10} t_{11} t_{12} Σ
$\ell_j + M_{10,j}$ 1.2 0.6 0.4 0.8 1.1 0.9 −∞ −∞
weight a_j 0.2318 0.1272 0.1042 0.1554 0.2097 0.1717 0 0 1
value $\mathbf{v}_j$ (1, 0) (0, 1) (1, 1) (−1, 1) (2, 0) (0, 2) – –
$a_j \mathbf{v}_j$ 0.2318 0.0000 0.1042 −0.1554 0.4195 0.0000 – – 0.6000
0.0000 0.1272 0.1042 0.1554 0.0000 0.3434 – – 0.7302
$a_j = \exp(\ell_j)/Z$, $Z = \Sigma_j \exp(\ell_j) = 14.32337$
$\mathbf{o}_{10} = \Sigma_j a_j \mathbf{v}_j = (0.600050, 0.730184)$
value space, $d_v = 2$
v_2
v_1
t_{10}
P_1
S_2
S_1
P_0
t_9
$\mathbf{o}_{10}$